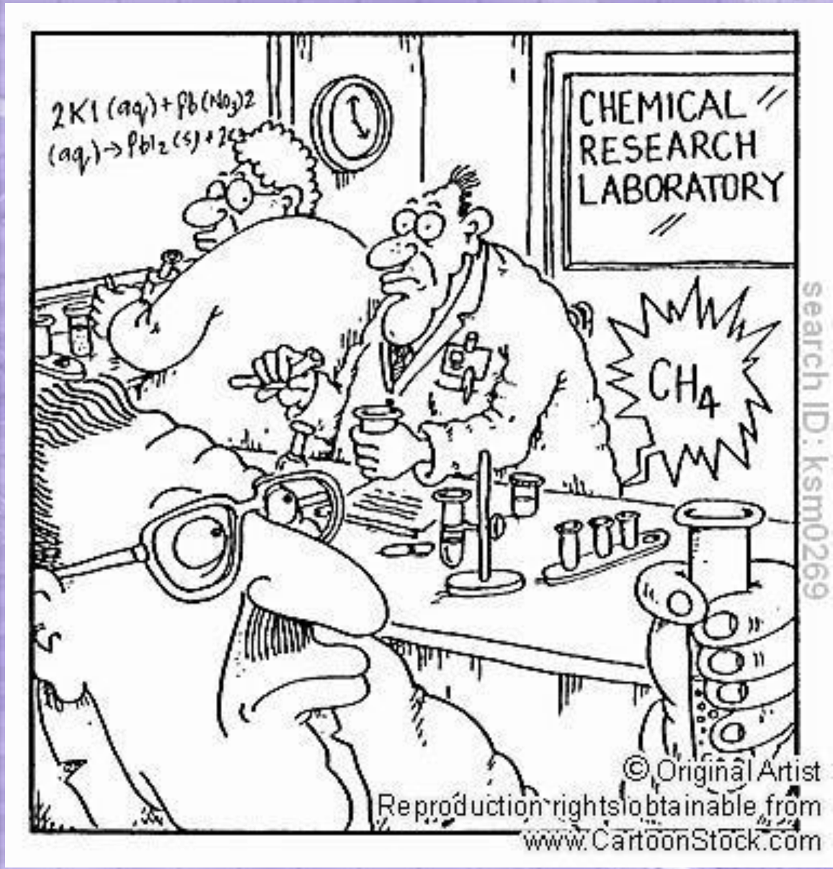
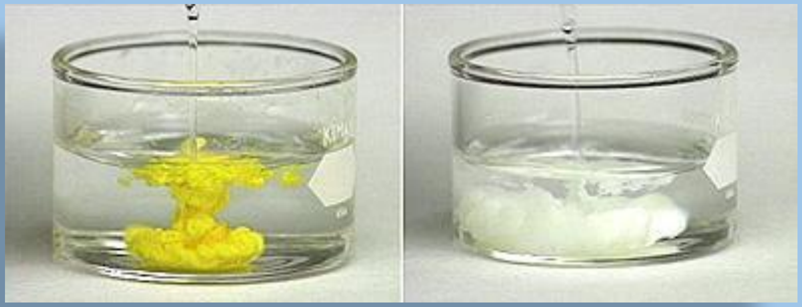
Ionic Reactions



Solubility

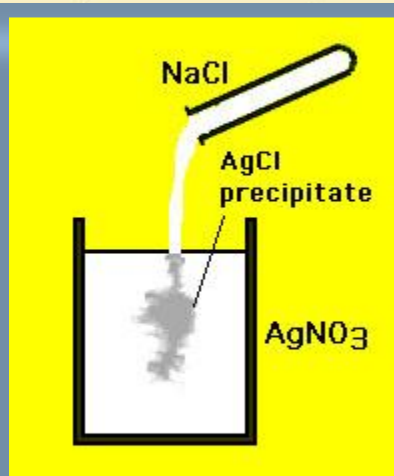
- =max. amount that dissolve/L of solvent at a specific temperature
- solubility $\geq$ 0.1 mol/L = **soluble**
- solubility < 0.1 mol/L = **insoluble** (highlight on table on your note)
- insoluble = **precipitate**

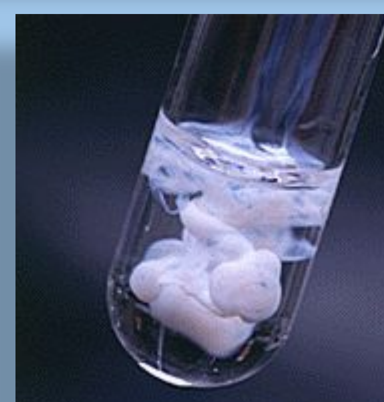


Example

- **$AgNO_{3(aq)}$ + $NaCl_{(aq)}$ → $AgCl_{(s)}$ + $NaNO_{3(aq)}$**

Cations		Anions						
		Cl^-, Br^-, I^-	S^{2-}	OH^-	SO_4^{2-}	CO_3^{2-}, PO_4^{3-}, SO_3^{2-}	$C_2H_3O_2^-$	NO_3^-
	High solubility (aq) $\geq$0.1 mol/L (at SATP)	most	Group 1, NH_4^+ Group 2	Group 1, NH_4^+ Sr^{2+}, Ba^{2+}, Tl^+	most	Group 1, NH_4^+	most	all
		All Group 1 compounds, including acids, and all ammonium compounds are assumed to have high solubility in water.						
	Low Solubility (s) <0.1 mol/L (at SATP)	Ag^+, Pb^{2+}, Tl^+, Hg_2^{2+}, (Hg^+), Cu^+	most	most	Ag^+, Pb^{2+}, Ca^{2+}, Ba^{2+}, Sr^{2+}, Ra^{2+}	most	Ag^+	none





Each person in group choose a different one of these reactions and write on your sheet.

Option 1:

$Mg(NO_3)_{2\ (aq)} + Na_2CO_{3\ (aq)} \rightarrow MgCO_{3\ (s)} + 2\ NaNO_{3\ (aq)}$

Option 2:

$2\ Cr(NO_3)_{3\ (aq)} + 3\ FeSO_{4\ (aq)} \rightarrow 3\ Fe(NO_3)_{2\ (aq)} + Cr_2(SO_4)_{3\ (aq)}$

Option 3:

$K_3PO_{4\ (aq)} + Al(NO_3)_{3\ (aq)} \rightarrow AlPO_{4\ (s)} + 3\ KNO_{3\ (aq)}$

Option 4:

$BeI_{2\ (aq)} + Cu_2SO_{4\ (aq)} \rightarrow BeSO_{4\ (aq)} + 2\ CuI_{\ (s)}$

STEP 1

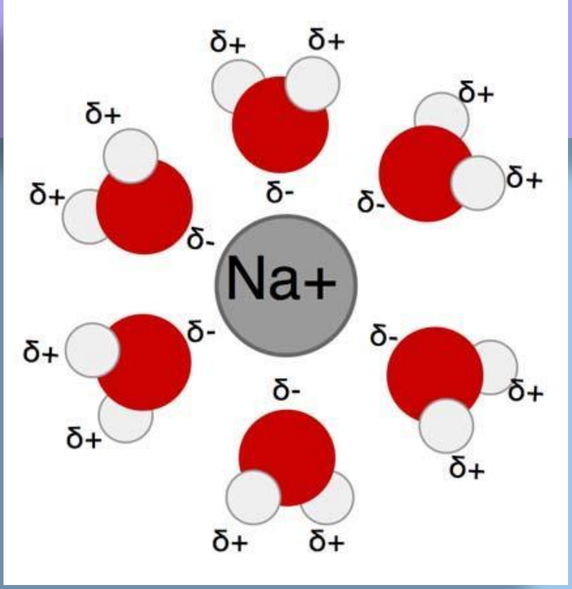
Review the next 3 slides, then complete for your reaction.

Check each other's work in your group.

Ionic Reactions

- to create an **ionic equation**, specify individual ions present
- e.g. dissolve $NaCl_{(s)}$ in water, it becomes:

$$Na^{+}_{(aq)} + Cl^{-}_{(aq)}$$



Ionic Reactions

- e.g. # 2 sodium hydroxide + hydrochloric acid:
- $NaOH_{(aq)} + HCl_{(aq)} \rightarrow NaCl_{(aq)} + H_2O_{(l)}$
- now identify ions (only aq!):
- $Na^{+}_{(aq)} + OH^{-}_{(aq)} + H^{+}_{(aq)} + Cl^{-}_{(aq)} \rightarrow Na^{+}_{(aq)} + Cl^{-}_{(aq)}$

  +

COUNT ALL IONS AND USE COEFFICENTS ONLY TO INDICATE NUMBER

examples:

2NaCl $_{(aq)}$ becomes 2Na $^{+}_{(aq)}$ + 2 Cl$^{-}_{(aq)}$

$MgCl_2$ $_{(aq)}$ becomes Mg $^{2+}_{(aq)}$ + 2 Cl$^{-}_{(aq)}$

What would 2$MgCl_2$ become?

WHY does AgCl NOT separate into ions?

Complete this separation to ions for ALL compounds in your reaction THAT ARE AQUEOUS and compare with group.

STEP 2

Review the next 3 slides then complete for your reaction.

Check each other's work in your group

Net Ionic Rxns

- now cancel ions appearing on both sides of the ionic equation
- called spectator ions
- as they are not really involved in reaction
- They begin the reaction

as dissolved ions and

are still the same

dissolved ion after

Net Ionic Rxns

- for our example:
- $\cancel{Na^+}_{(aq)} + OH^-_{(aq)} + H^+_{(aq)} + \cancel{Cl^-}_{(aq)} \rightarrow Na^+_{(aq)} + Cl^-_{(aq)} + H_2O_{(l)}$

- $OH^-_{(aq)} + H^+_{(aq)} \rightarrow H_2O_{(l)}$
- or base + acid → water

note that only (aq) species become ions and may become spectator ions

NOTE:

Coefficients AND charges AND states must all be identical to fully cancel out.

Complete Step 2 and compare in your groups

STEP 3

Just rewrite what remains (everything NOT crossed out on either side). See next slide for example.

Compare with your group and with other groups.

Net Ionic Rxns

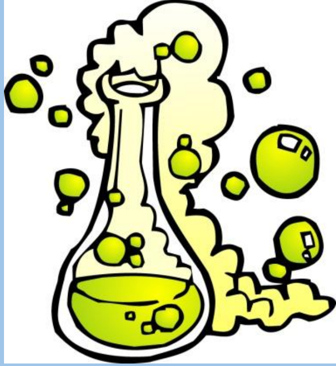
- consider the original example
- $AgNO_{3(aq)} + NaCl_{(aq)} \rightarrow AgCl_{(s)} + NaNO_{3(aq)}$
- total ionic:
- $Ag^{+}_{(aq)} + \cancel{NO}_3{}^{-}{}_{(aq)} + \cancel{Na}^{+}_{(aq)} + Cl^{-}_{(aq)} \rightarrow AgCl_{(s)} + Na^{+}_{(aq)} + NO_3{}^{-}{}_{(aq)}$
- net ionic:

$$Ag^{+}_{(aq)} + Cl^{-}_{(aq)} \rightarrow AgCl_{(s)}$$

Now lets try this one:

Ni(NO3)3 (aq) + 3 KBr (aq) NiBr3 (aq) + 3 KNO3 (aq)

Rxn Drive



- net ionic equation also reflects one reason the reaction moved forward